

RPM Measurement: Sensors and Techniques

Low-cost, high-value sensor options are available for revolutions-per-minute (RPM) measurement applications. Most of these sensors provide simple, reliable and inexpensive transducers for highly sophisticated control systems. Take a look at different types of tachometers for RPM measurement, advanced sensor modules, and frequency measurement and period measurement methods



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Detecting the stoppage of a rotating shaft (zero-speed detection) is significant in many applications, for example, to determine whether a conveyor has failed. Revolutions per minute (RPM) measurement is crucial when controlling or monitoring the speed of motors, conveyors, turbines, etc. RPM is a measure of the frequency of a rotation. It annotates the number of full rotations completed in one minute around a fixed axis. It is used as a measure of rotational speed of a mechanical component. Advanced electronics technology now offers low-cost, high-value sensor options for RPM measurement applications. Most of these sensors provide simple, reliable and inexpensive transducers for highly sophisticated control systems.

RPM sensors

A sensor is essential to sense shaft speed. Typically, devices used for this purpose are shaft (rotary type) encoders, photoelectric (optical type) sensors and magnetic rotational speed (proximity type) sensors. All of these sensors send speed data in the form of electrical pulses. Shaft encoders offer a high resolution

of typically 1-5000 pulses per revolution (PPR) and clearly defined, symmetrical pulses. Proximity sensors provide medium (or low) resolution sensing, depending on the number of pulses measured per revolution. Photoelectric sensors usually sense a reflective target on the rotating shaft. Magnetic rotational speed sensors use various magnetic proximity measuring principles to monitor the speeds of machine components in a range between 0 and 30,000 rpm.

In principle, RPM sensors convert mechanical motion into electric pulses with or without direct contact when positioned near a turning rotor, gear, shaft or other regularly moving device. The resultant output signals are then fed to a digital counter, totaliser, tachometer, or other monitoring and control device.

Tachometers

The word 'tachometer' is derived from the Greek words 'tachos,' meaning 'speed,' and 'metron,' meaning 'to measure.' Tachometer is a device designed to measure rotational or surface speed either in contact or non-contact way. Tachometers are widely used to measure rotational speed of motors, conveyor belts, and other moving or rotating systems. RPM measurement with a tachometer can simply be divided into three popular methods:

Mechanical. In this method, revolutions in the mechanical sensor are electronically analysed in the instrument. This good old method is still commonly used for low resolutions between 20 and 20,000 rpm. Mechanical tachometers utilise the fact that the centrifugal force on a rotating mass depends on the speed of rotation.

Optical. Here rotation is transmitted to the measuring instrument via an infrared (IR) light beam/laser beam coming from the instrument, which is then reflected by a reflective tape on the object.

As explained, traditional tachometers require physical contact between the instruments and the objects being monitored. Laser tachometer is a powerful choice where this type of direct-contact type measurement is not workable for technical or safety reasons. Laser tachometers work by pulsing a laser beam against the rotating element. The rotating element will have one reflective spot, and the